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Deep Learning

DL101 | Suggested sequence · Semester 4 | Student-authored study guide

How to use these notes

Neural-network building blocks, backpropagation, optimization, regularization, and careful evaluation. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Neurons and activations

A layer computes weighted sums and a nonlinear activation. Without nonlinearities, stacked linear layers collapse to a single linear transformation. Output activation and loss should match the task.

Worked example

```
A binary classifier commonly uses a logit with binary cross-entropy; a multi-class classifier
uses a class-score vector.
```

Check your understanding

- Why add nonlinear activation?
- What do weights and biases control?

MODULE 02 · Loss, gradients, and training

Training minimizes a loss using gradients computed by backpropagation. Mini-batches estimate gradients efficiently. Track training and validation loss to detect optimization problems and overfitting.

Worked example

```
A decreasing training loss with rising validation loss is a signal to examine generalization.
```

Check your understanding

- What is an epoch?
- What can a validation curve reveal?

MODULE 03 · Generalization and safe evaluation

Regularization, early stopping, data augmentation, and appropriate model size can help generalization. Split by person/time/group when repeated records would otherwise leak identity or future information.

Worked example

```
For patient records, split by patient, not individual visit, to prevent the same person
appearing on both sides.
```

Check your understanding

- What is overfitting?
- Why can random row splitting be misleading?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.